Functions & Transformations

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Reading a function graph**

The graph of $y = f(x) = x^2 - 4$ is shown.

a State the zeros of f. $x = -2$ and $x = 2$ (where the curve crosses the x-axis).

b State the domain and range of f. Domain: all real numbers. Range: $y \geq -4$.

(a) $x = -2$ and $x = 2$ (where the curve crosses the x-axis).; (b) Domain: all real numbers. Range: $y \geq -4$.

2 **[Foundational]** • **Target: Evaluating functions**

If $f(x) = 3x - 4$, find $f(5)$.

$f(5) = 3(5) - 4 = 11$.

3 **[Foundational]** • **Target: Evaluating functions**

If $f(x) = x^2 - 2x + 1$, find $f(-2)$.

$f(-2) = 4 + 4 + 1 = 9$.

4 **[Foundational]** • **Target: Evaluating functions**

If $g(x) = \sqrt{x + 5}$, find $g(4)$.

$g(4) = \sqrt{9} = 3$.

5 **[Foundational]** • **Target: Evaluating functions**

If $h(x) = \frac{2x}{x - 1}$, find $h(3)$.

$h(3) = \frac{6}{2} = 3$.

6 **[Foundational]** • **Target: Domain and range**

State the domain of $f(x) = \sqrt{x - 3}$.

$x - 3 \geq 0$, so the domain is $x \geq 3$.

7 **[Foundational]** • **Target: Domain and range**

State the domain of $f(x) = \frac{1}{x - 5}$.

All real numbers except $x = 5$ (where the denominator is zero).

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range

State the domain of $f(x) = \sqrt{7 - x}$.

$7 - x \geq 0$, so the domain is $x \leq 7$.

9 [Intermediate] • Target: Domain and range

State the range of $f(x) = x^2 + 2$.

Since $x^2 \geq 0$ for all real x, the range is $f(x) \geq 2$.

10 [Intermediate] • Target: Transformations: shifts

The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.

$y = (x - 3)^2 - 2$.

11 [Intermediate] • Target: Transformations: shifts

The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.

$y = |x + 4| + 1$.

12 [Intermediate] • Target: Transformations: shifts

Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.

A shift 2 units left and 5 units down.

13 [Intermediate] • Target: Transformations: reflections and stretches

The graph of $y = x^2$ is reflected over the x-axis and stretched vertically by a factor of 3. Write the new equation.

$y = -3x^2$.

14 [Intermediate] • Target: Transformations: reflections and stretches

Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.

A vertical stretch by a factor of 2.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Transformations: reflections and stretches • Cross-Lesson: 3.01 Sequences & Series
Describe the transformation that takes $y = x^2$ to $y = -(x-1)^2 + 4$.

A reflection over the x-axis, then a shift 1 unit right and 4 units up.

16 [Challenging] • Target: Composite functions • Cross-Lesson: 4.01 Trigonometry
Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.

$g(2) = 4 - 3 = 1$. $f(1) = 2(1) + 1 = 3$.

17 [Challenging] • Target: Composite functions • Cross-Lesson: 5.01 Geometry & Coordinate Geometry
Using the same f and g, find $(g \circ f)(2)$.

$f(2) = 5$. $g(5) = 25 - 3 = 22$.

18 [Challenging] • Target: Composite functions • Cross-Lesson: 6.01 Limits & Derivatives
Using the same f and g, find a general formula for $(f \circ g)(x)$.

$f(g(x)) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5$.

19 [Challenging] • Target: Composite functions • Cross-Lesson: 7.01 Integration & Applications
Using the same f and g, find a general formula for $(g \circ f)(x)$.

$g(f(x)) = (2x + 1)^2 - 3 = 4x^2 + 4x + 1 - 3 = 4x^2 + 4x - 2$.

20 [Challenging] • Target: Inverse functions • Cross-Lesson: 8.01 Exponentials & Logarithms
Find the inverse of $f(x) = 3x - 2$.

$y = 3x - 2 \Rightarrow x = 3y - 2 \Rightarrow y = \frac{x+2}{3}$. So $f^{-1}(x) = \frac{x+2}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Domain and range
The graph shows $f(x)$ defined only where plotted. State its domain.

a. $[-3, 3]$ ✓

b. All real numbers

c. $[-4, 4]$

d. $[0, 3]$

The curve is only plotted for x between -3 and 3.

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22 [Foundational] • Target: MCQ — Even and odd functions

The graph shown is symmetric about the y-axis. Is the function even, odd, or neither?

a. Odd

b. Neither

c. Cannot be determined

d. **Even** ✓

Symmetry about the y-axis is exactly the definition of an even function.

23 [Foundational] • Target: MCQ — Reading a function graph

The graph shows $y = f(x) = x^2 - 9$. Find $f(3)$ by reading the graph.

a. -9

b. 6

c. 9

d. 0 ✓

The curve crosses the x-axis at $x = 3$, so $f(3) = 0$.

24 [Foundational] • Target: MCQ — Reading a function graph

Using the same graph, find the values of x where $f(x) = 0$ (the x-intercepts).

a. $x = 0$ and $x = 9$

b. $x = -9$ and $x = 9$

c. $x = 3$ only

d. $x = -3$ **and** $x = 3$ ✓

The curve crosses the x-axis at $x = -3$ and $x = 3$.

25 [Foundational] • Target: MCQ — Reading a function graph

Using the same graph, find the y-intercept.

a. -3

b. 9

c. 0

d. -9 ✓

At $x = 0$, the curve crosses the y-axis at $y = -9$.

26 [Foundational] • Target: MCQ — Evaluating functions

If $f(x) = 4x - 7$, find $f(6)$.

a. 17 ✓

b. 24

c. 31

d. 10

$4(6) - 7 = 24 - 7 = 17$.

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27 [Foundational] • Target: MCQ — Evaluating functions

If $f(x) = 2x^2 + 1$, find $f(3) - f(1)$.

a. 18

b. 8

c. 16 ✓

d. 22

$f(3) = 19$, $f(1) = 3$; $19 - 3 = 16$.

28 [Foundational] • Target: MCQ — Domain and range

State the range of $f(x) = x^2 + 5$.

a. $y > 5$

b. All real numbers

c. $y \leq 5$

d. $y \geq 5$ ✓

Since $x^2 \geq 0$, the minimum value of f is 5: range $y \geq 5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — Transformations: reflections and stretches

The graph of $y = x^2$ is reflected over the x-axis and stretched vertically by a factor of 2. Write the new equation.

a. $y = -\frac{1}{2}x^2$

b. $y = (-2x)^2$

c. $y = -2x^2$ ✓

d. $y = 2x^2$

Reflect over x-axis: negate; stretch by 2: multiply by 2.

30 [Intermediate] • Target: MCQ — Composite functions

Let $f(x) = 3x + 2$ and $g(x) = x^2 - 1$. Find $(f \circ g)(3)$.

a. 26 ✓

b. 23

c. 11

d. 29

$g(3) = 8$; $f(8) = 3(8) + 2 = 26$.

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31 [Intermediate] • Target: MCQ — Inverse functions

Show that $f(x) = 3x + 9$ and $g(x) = \frac{x-9}{3}$ are inverses by computing $f(g(x))$.

a. x ✓

b. $x + 9$

c. $3x$

d. $x - 9$

$f(g(x)) = 3\left(\frac{x-9}{3}\right) + 9 = x - 9 + 9 = x$.

32 [Intermediate] • Target: MCQ — Piecewise functions

Using the same piecewise function, find $f(2)$.

a. 3 ✓

b. 4

c. 0

d. 5

$x = 2 \geq 2$: $f(2) = 2(2) - 1 = 3$.

33 [Intermediate] • Target: MCQ — Even and odd functions

Determine whether $f(x) = x^2 + 3x$ is even, odd, or neither.

a. Both even and odd

b. **Neither** ✓

c. Odd

d. Even

$f(-x) = x^2 - 3x$, which is neither $f(x)$ nor $-f(x)$: neither.

34 [Intermediate] • Target: MCQ — Increasing, decreasing, and average rate of change

Find the average rate of change of $f(x) = x^3$ from $x = 1$ to $x = 3$.

a. 8

b. 26

c. 9

d. 13 ✓

$f(3) = 27$, $f(1) = 1$; rate $= \frac{26}{2} = 13$.

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35 [Intermediate] • Target: MCQ — Word problems: functions in context

A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 300$. Find the profit when 15 units are sold.

a. 450 **dollars** ✓

b. 150 dollars

c. 600 dollars

d. 900 dollars

$P(15) = -2(225) + 1{,}200 - 300 = 450$ dollars.

36 [Intermediate] • Target: MCQ — Transformations: reflections and stretches

If $f(x) = -3x^2 - 5x + 9$, find $f(-3)$.

a. 0

b. -6

c. -3 ✓

d. 3

$f(-3) = -3(-3)^2 + 15 + 9 = -3$.

37 [Intermediate] • Target: MCQ — Composite functions

If $f(x) = x^2 + 8x - 1$, find $f(-5)$.

a. -19

b. -16 ✓

c. -13

d. 16

$f(-5) = (-5)^2 - 40 - 1 = -16$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Inverse functions • Cross-Lesson: 6.01 Limits & Derivatives

If $f(x) = x^2 + 6x + 9$, find $f(-5)$.

a. 4 ✓

b. 1

c. 7

d. -4

$f(-5) = (-5)^2 - 30 + 9 = 4$.

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39 [Challenging] • Target: MCQ — Piecewise functions • Cross-Lesson: 7.01 Integration & Applications

If $f(x) = x^2 - 2x - 2$, find $f(-2)$.

a. 9

b. -6

c. 3

d. 6 ✓

$f(-2) = (-2)^2 + 4 - 2 = 6.$

40 [Challenging] • Target: MCQ — Even and odd functions • Cross-Lesson: 8.01 Exponentials & Logarithms

State the domain of $f(x) = \sqrt{3x + 6}$.

a. $x \geq -2$ ✓

b. $x \geq -4$

c. $x \leq -2$

d. $x \geq 0$

$3x + 6 \geq 0 \Rightarrow x \geq -2.$

41 [Challenging] • Target: MCQ — Increasing, decreasing, and average rate of change • Cross-Lesson: 9.01 Statistics & Probability

State the domain of $f(x) = \sqrt{x - 2}$.

a. $x \leq 2$

b. $x \geq 4$

c. $x \geq 0$

d. $x \geq 2$ ✓

$x - 2 \geq 0 \Rightarrow x \geq 2.$

42 [Challenging] • Target: MCQ — Operations on functions • Cross-Lesson: 1.01 Algebra & Equations

State the domain of $f(x) = \sqrt{-2x + 14}$.

a. $x \leq 7$ ✓

b. $x \leq 5$

c. $x \geq 7$

d. $x \leq 9$

$-2x + 14 \leq 0 \Rightarrow x \leq 7.$

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43 [Challenging] • Target: MCQ — Reading a function graph • Cross-Lesson: 3.01 Sequences & Series

Let $f(x) = -x - 2$ and $g(x) = -x^2 + 3$. Find $(f \circ g)(1)$.

a. -4 ✓

b. -7

c. -1

d. 4

$g(1) = -(1)^2 + 3 = 2$. $f(2) = -(2) - 2 = -4$.

44 [Challenging] • Target: MCQ — Word problems: functions in context • Cross-Lesson: 4.01 Trigonometry

Find the inverse of $f(x) = 5x + 1$.

a. $\frac{x-1}{5}$ ✓

b. $\frac{x-1}{-5}$

c. $\frac{x+1}{5}$

d. $5x + 1$

$y = 5x + 1 \Rightarrow x = 5y + 1 \Rightarrow y = \frac{x-1}{5}$. So $f^{-1}(x) = \frac{x-1}{5}$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

The graph of $y = |x|$ is shifted 7 units left and 3 units down. Write the new equation.

a. $y = |x + 7| + 3$

b. $y = |x - 7| - 3$

c. $y = |x + 7| - 3$ ✓

d. $y = |x - 7| + 3$

$y = |x + 7| - 3$.

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